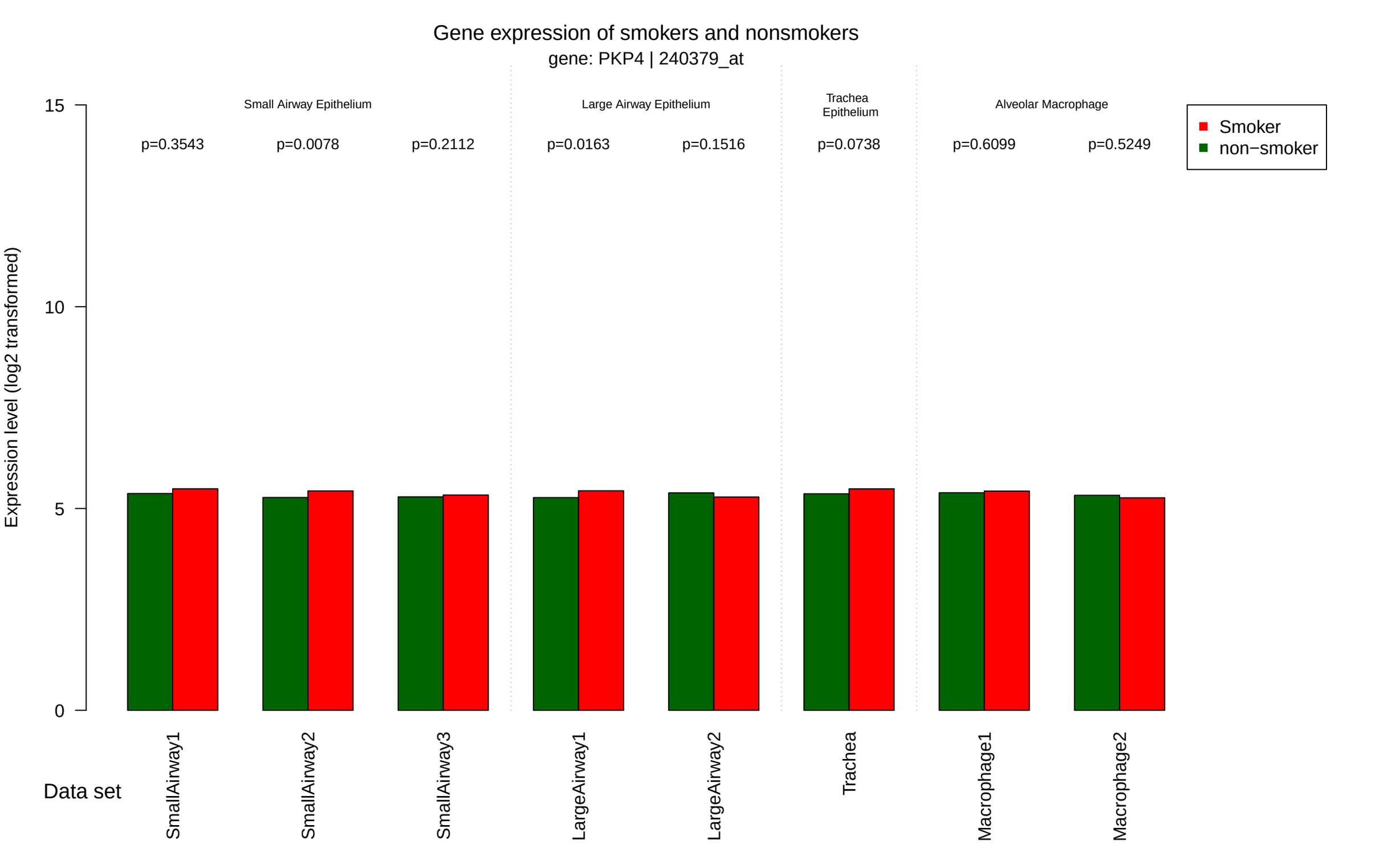
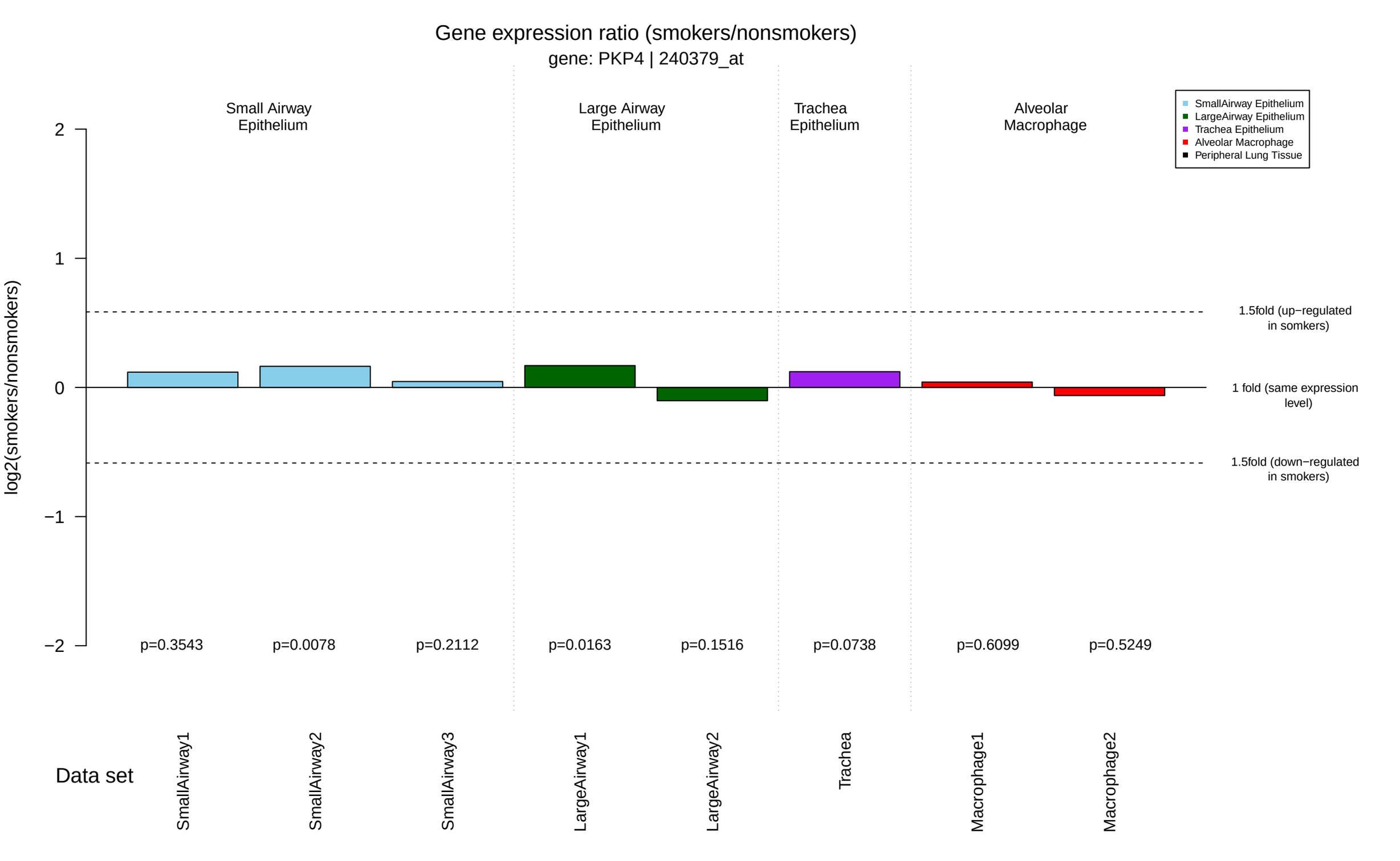
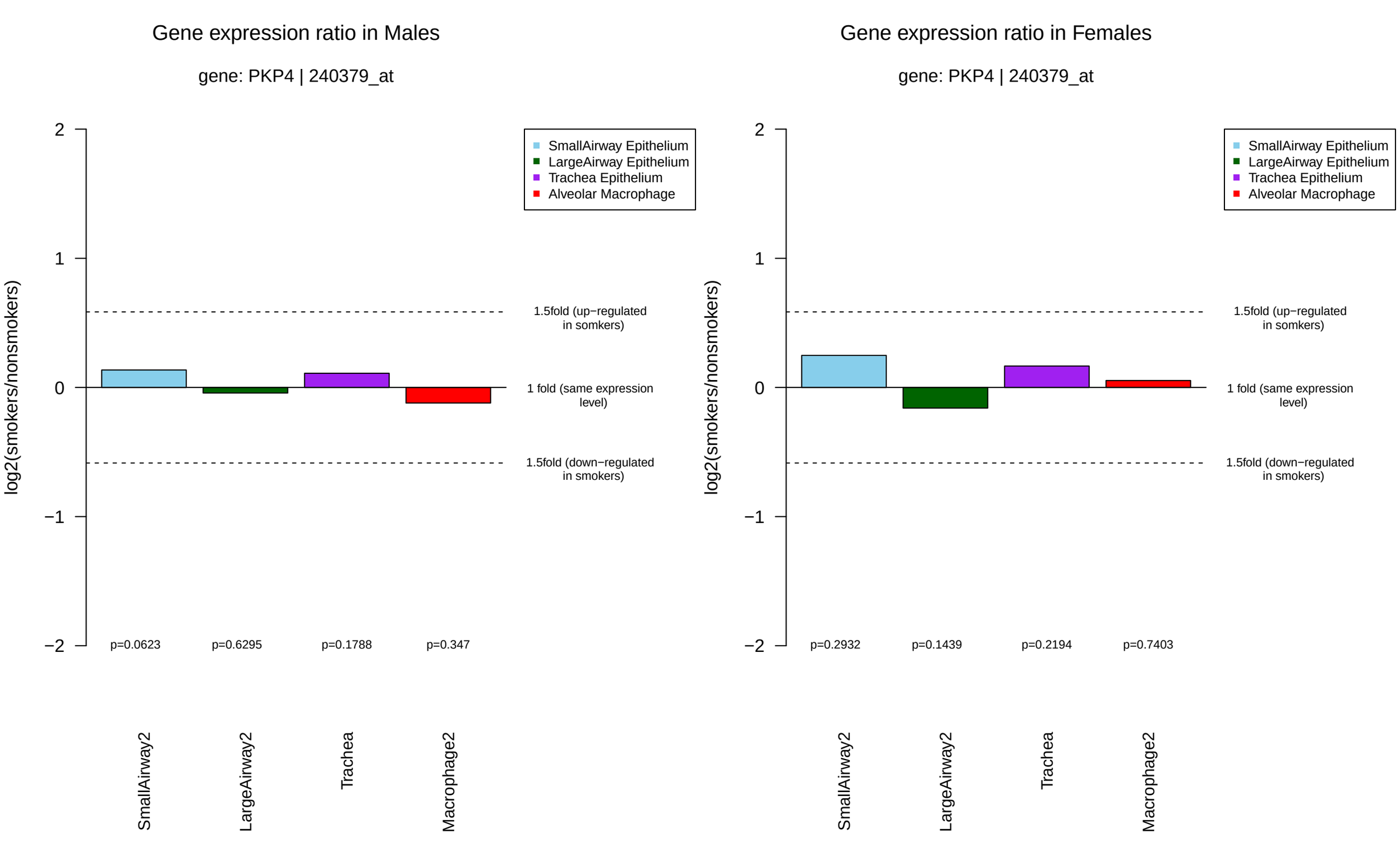
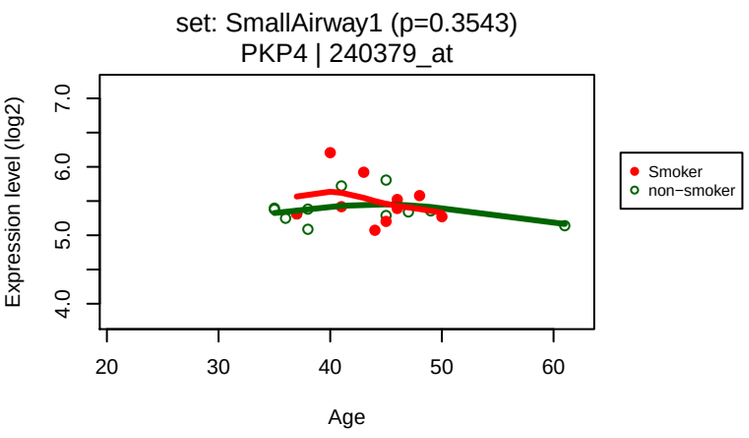
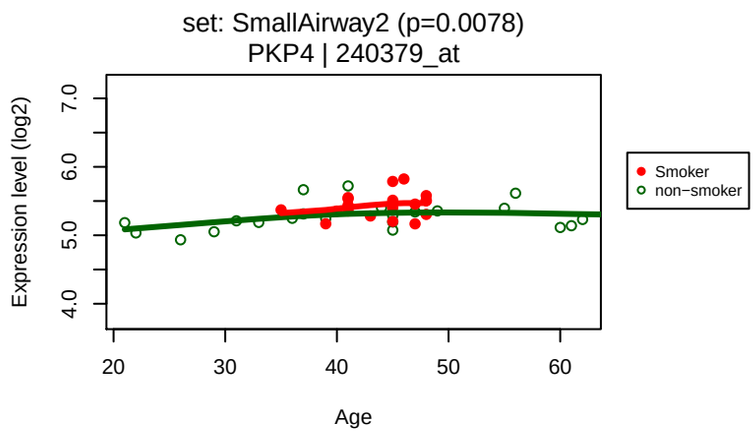
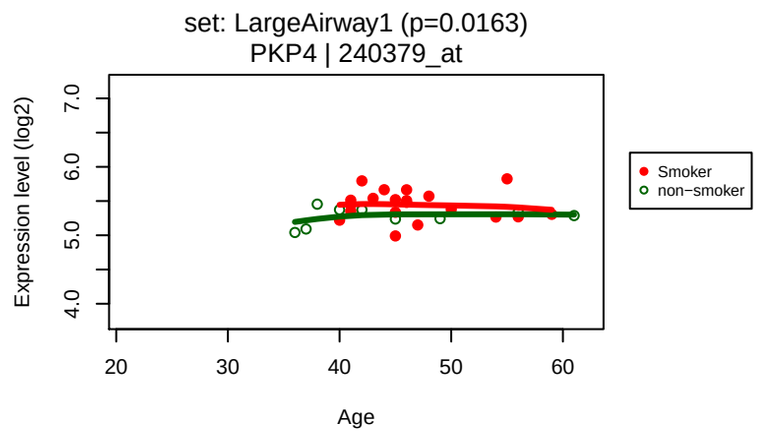

Gene expression of smokers and nonsmokers
gene: PKP4 | 240379_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3543
p=0.0078
p=0.2112
p=0.0163
p=0.1516
p=0.0738
p=0.6099
p=0.5249
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

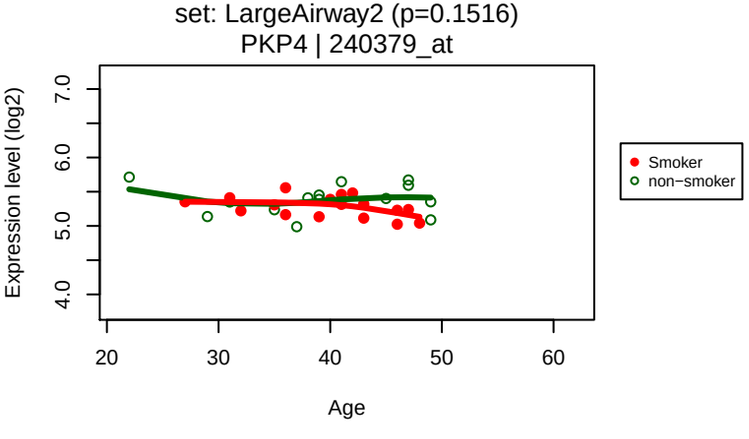

Gene expression ratio (smokers/nonsmokers)
gene: PKP4 | 240379_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3543
p=0.0078
p=0.2112
p=0.0163
p=0.1516
p=0.0738
p=0.6099
p=0.5249
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

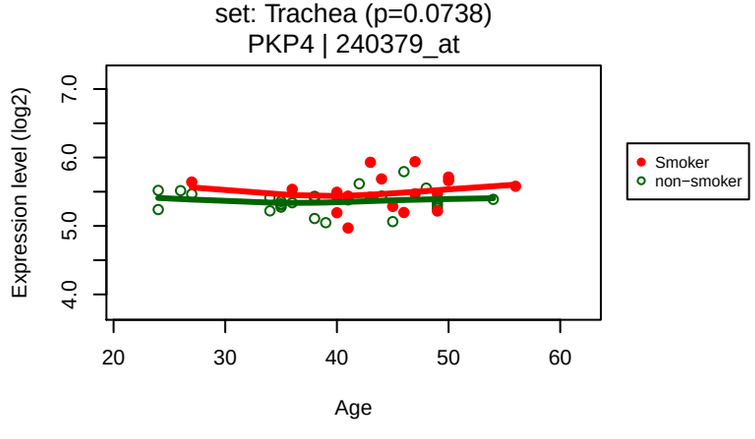

Gene expression ratio in Males
gene: PKP4 | 240379_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0623
p=0.6295
p=0.1788
p=0.347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PKP4 | 240379_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2932
p=0.1439
p=0.2194
p=0.7403
SmallAirway2
LargeAirway2
Trachea
Macrophage2

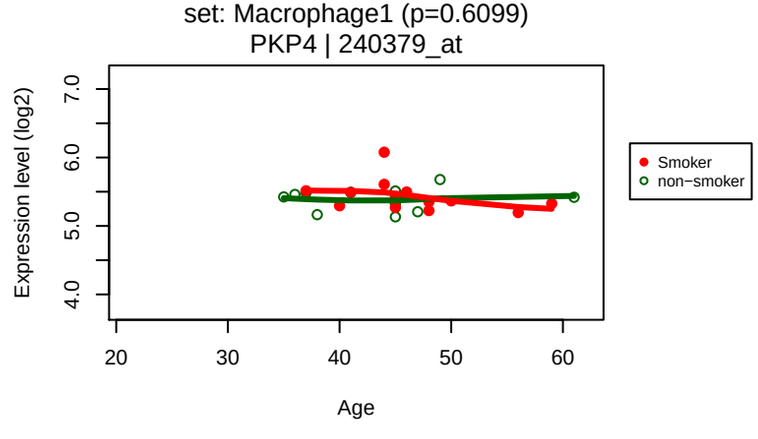

set: SmallAirway1 (p=0.3543)
PKP4 | 240379_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

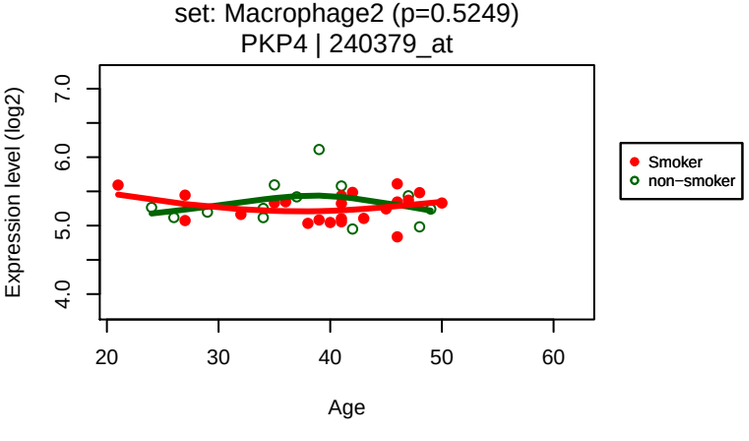

set: SmallAirway2 (p=0.0078)
PKP4 | 240379_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

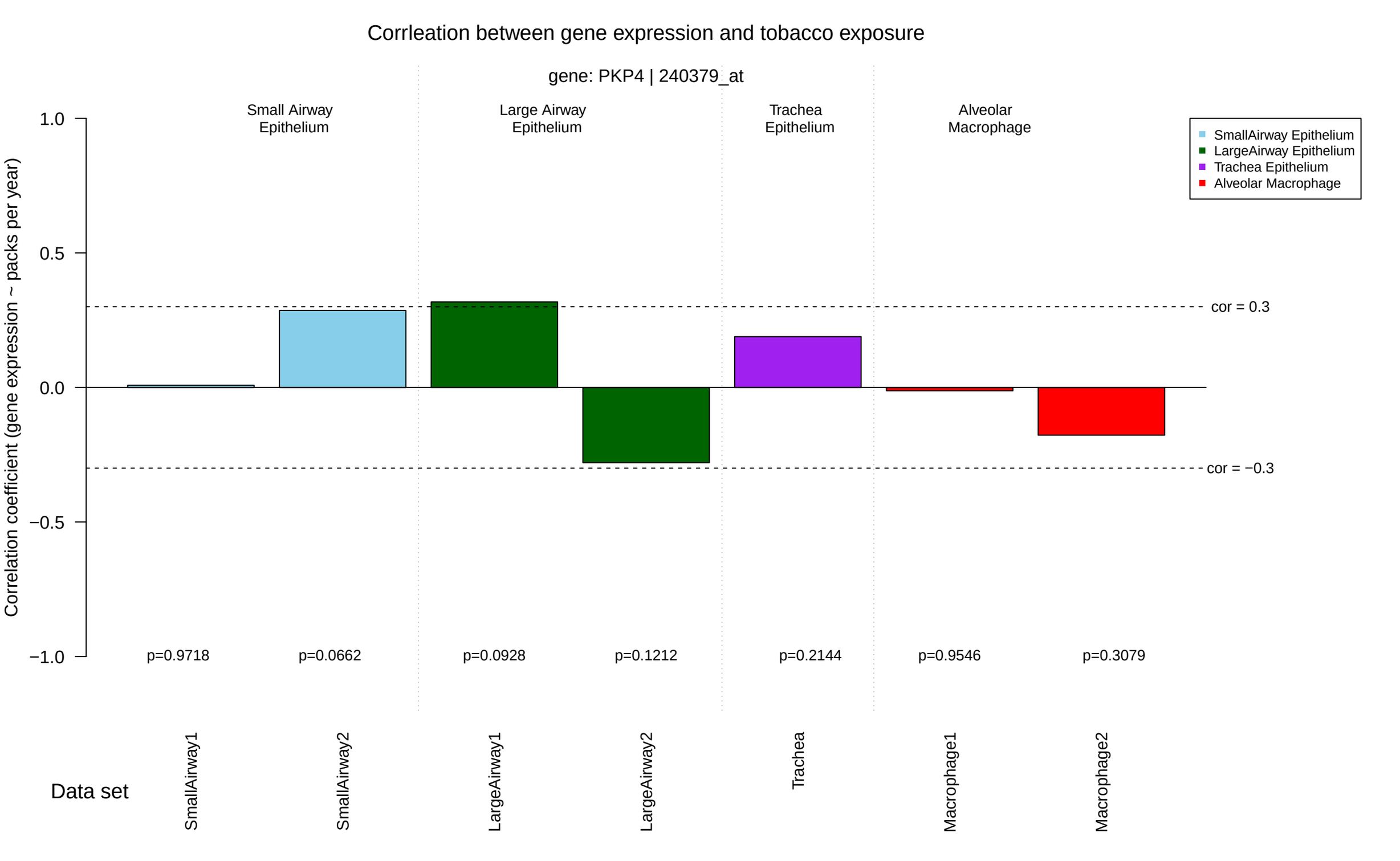

set: LargeAirway1 (p=0.0163)
PKP4 | 240379_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1516)
PKP4 | 240379_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0738)
PKP4 | 240379_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6099)
PKP4 | 240379_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5249)
PKP4 | 240379_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PKP4 | 240379_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9718
p=0.0662
p=0.0928
p=0.1212
p=0.2144
p=0.9546
p=0.3079
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2